

Natural language statement

Domain

GroundTruth

Definition. A topological space X is called a realcompact space if X is a Tychonoff space and there is no Tychonoff space X which satisfies: (RC1) there exists a homeomorphic embedding $r: X \rightarrow X$ such that $r(X) \neq \text{closure}(r(X)) = X$; (RC2) for every continuous real-valued function $f: X \rightarrow \mathbb{R}$ there exists a continuous function $f: X \rightarrow \mathbb{R}$ such that $fr = f$.

3.11.16. Theorem. For every Tychonoff space X there exists exactly one (up to a homeomorphism) realcompact space $\text{nu}X$ which satisfies: (i) there exists a homeomorphic embedding $\text{nu}: X \rightarrow \text{nu}X$ such that $\text{closure}(\text{nu}(X)) = \text{nu}X$; (ii) for every continuous real-valued function $f: X \rightarrow \mathbb{R}$ there exists a continuous function $f^*\text{nu}: \text{nu}X \rightarrow \mathbb{R}$ such that $f^*\text{nu} \circ \text{nu} = f$. The space $\text{nu}X$ also satisfies: (iii) for every continuous mapping $f: X \rightarrow Y$ of X to a realcompact space Y there exists a continuous mapping $f^*\text{nu}: \text{nu}X \rightarrow Y$ such that $f^*\text{nu} \circ \text{nu} = f$.

Topology

```
def IsRealcompact (X : Type u) [tX : TopologicalSpace X] : Prop :=
  T3Space X /\ not exists (Y : Type u) (tY : TopologicalSpace Y),
    @T3Space Y tY /\ exists r : X -> Y,
      @IsEmbedding X Y tX tY r /\ range r != closure (range r) /\
        closure (range r) = univ /\
          forall f : X -> R, Continuous f ->
            exists g : Y -> R, @Continuous Y R tY _ g /\ g comp r = f

theorem engelking_3_11_16_hewitt_realcompactification
  {X : Type u} [TopologicalSpace X] [T3Space X] :
  exists (Y : Type u) (tY : TopologicalSpace Y) (nu : X -> Y),
    @IsRealcompact Y tY /\ @IsEmbedding X Y _ tY nu /\ closure (range nu)
      = univ /\
        (forall f : X -> R, Continuous f ->
          exists g : Y -> R, @Continuous Y R tY _ g /\ g comp nu = f) /\
        (forall (Z : Type u) (tZ : TopologicalSpace Z), @IsRealcompact Z tZ
          ->
            forall f : X -> Z, @Continuous X Z _ tZ f ->
              exists g : Y -> Z, @Continuous Y Z tY tZ g /\ g comp nu = f) /\
        forall (Z : Type u) (tZ : TopologicalSpace Z) (zeta : X -> Z),
          @IsRealcompact Z tZ -> @IsEmbedding X Z _ tZ zeta -> closure
            (range zeta) = univ ->
              (forall f : X -> R, Continuous f ->
                exists g : Z -> R, @Continuous Z R tZ _ g /\ g comp zeta = f) ->
                exists h : @Homeomorph Y Z tY tZ, h comp nu = zeta := by
sorry
```

Definition. A topological space X is called countably paracompact if X is a Hausdorff space and every countable open cover of X has a locally finite open refinement. 5.2.8. Theorem. A topological space X is normal and countably paracompact if and only if the Cartesian product $X \times I$ of X and the closed unit interval I is normal.

Topology

```
def IsOpenCover {X : Type u} [TopologicalSpace X] {iota : Type v}
  (U : iota -> Set X) : Prop :=
  (forall i, IsOpen (U i)) /\ union i, U i = univ

def Refines {X : Type u} {iota : Type v} {kappa : Type w} (V : kappa -> Set
  X)
  (U : iota -> Set X) : Prop :=
  forall j, exists i, V j subset U i

def IsCountablyParacompact (X : Type u) [TopologicalSpace X] : Prop :=
  T2Space X /\
    forall (iota : Type v) (_ : Countable iota) (U : iota -> Set X),
      IsOpenCover U ->
        exists (kappa : Type v) (V : kappa -> Set X), IsOpenCover V /\
          Refines V U /\ LocallyFinite V

theorem engelking_5_2_8_normal_product_interval
  {X : Type u} [TopologicalSpace X] :
  (NormalSpace X /\ IsCountablyParacompact X) <=>
    NormalSpace (X x Set.Icc (0 : R) 1) := by
sorry
```

Definitions. A family $\{A_s\}_{s \in S}$ is star-finite (star-countable) if for every s_0 in S the set $\{s \in S : A_s \text{ inter } A_{s_0} \neq \emptyset\}$ is finite (countable). A topological space X is called strongly paracompact if X is a Hausdorff space and every open cover of X has a star-finite open refinement. 5.3.10. Theorem. For every regular space X the following conditions are equivalent: (i) X is strongly paracompact; (ii) every open cover of X has a closed refinement which is both locally finite and star-finite; (iii) every open cover of X has a closed refinement which is both locally finite and star-countable; (iv) every open cover of X has a star-countable open refinement.

Topology

```
def IsOpenCover {X : Type u} [TopologicalSpace X] {iota : Type v}
  (U : iota -> Set X) : Prop :=
  (forall i, IsOpen (U i)) /\ union i, U i = univ

def Refines {X : Type u} {iota : Type v} {kappa : Type w} (V : kappa -> Set
  X)
  (U : iota -> Set X) : Prop :=
  forall j, exists i, V j subset U i

def IsClosedCover {X : Type u} [TopologicalSpace X] {iota : Type v}
  (F : iota -> Set X) : Prop :=
  (forall i, IsClosed (F i)) /\ union i, F i = univ

def IsStarFiniteFamily {X : Type u} {iota : Type v} (A : iota -> Set X) :
  Prop :=
  forall i, {j | (A i inter A j).Nonempty}.Finite

def IsStarCountableFamily {X : Type u} {iota : Type v} (A : iota -> Set X)
  : Prop :=
  forall i, {j | (A i inter A j).Nonempty}.Countable

def IsStronglyParacompact (X : Type u) [TopologicalSpace X] : Prop :=
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		<pre> T2Space X /\ forall (iota : Type v) (U : iota -> Set X), IsOpenCover U -> exists (kappa : Type v) (V : kappa -> Set X), IsOpenCover V /\ Refines V U /\ IsStarFiniteFamily V theorem engelking_5_3_10_strong_paracompactness {X : Type u} [TopologicalSpace X] [RegularSpace X] [T1Space X] : let closedLocallyFiniteStarFinite := forall (iota : Type v) (U : iota -> Set X), IsOpenCover U -> exists (kappa : Type v) (F : kappa -> Set X), IsClosedCover F /\ Refines F U /\ LocallyFinite F /\ IsStarFiniteFamily F let closedLocallyFiniteStarCountable := forall (iota : Type v) (U : iota -> Set X), IsOpenCover U -> exists (kappa : Type v) (F : kappa -> Set X), IsClosedCover F /\ Refines F U /\ LocallyFinite F /\ IsStarCountableFamily F let starCountableOpen := forall (iota : Type v) (U : iota -> Set X), IsOpenCover U -> exists (kappa : Type v) (V : kappa -> Set X), IsOpenCover V /\ Refines V U /\ IsStarCountableFamily V List.TFAE [IsStronglyParacompact X, closedLocallyFiniteStarFinite, closedLocallyFiniteStarCountable, starCountableOpen] := by sorry </pre>
4.4.1. The Stone Theorem. Every open cover of a metrizable space has an open refinement which is both locally finite and sigma-discrete.	Topology	<pre> def IsDiscreteFamily {X : Type u} [TopologicalSpace X] {iota : Type v} (U : iota -> Set X) : Prop := forall x : X, exists V in nhds x, {i : iota (V inter U i).Nonempty}.Subsingleton def IsOpenCover {X : Type u} [TopologicalSpace X] {iota : Type v} (U : iota -> Set X) : Prop := (forall i, IsOpen (U i)) /\ union i, U i = univ def Refines {X : Type u} {iota : Type v} {kappa : Type w} (V : kappa -> Set X) (U : iota -> Set X) : Prop := forall j, exists i, V j subset U i theorem engelking_4_4_1_stone_open_refinement {X : Type u} [TopologicalSpace X] [MetrizableSpace X] : forall (iota : Type v) (U : iota -> Set X), IsOpenCover U -> exists (kappa : Type w) (V : kappa -> Set X) (level : kappa -> N), IsOpenCover V /\ Refines V U /\ LocallyFinite V /\ forall n, IsDiscreteFamily fun j : {j : kappa // level j = n} => V j := by sorry </pre>
4.4.7. The Nagata-Smirnov Metrization Theorem. A topological space is metrizable if and only if it is regular and has a sigma-locally finite base.	Topology	<pre> def HasSigmaLocallyFiniteBase (X : Type u) [TopologicalSpace X] : Prop := exists (iota : N -> Type v) (B : forall n, iota n -> Set X), IsTopologicalBasis {V : Set X exists n i, B n i = V} /\ forall n, LocallyFinite (B n) theorem engelking_4_4_7_nagata-smirnov_metrization {X : Type u} [TopologicalSpace X] [T1Space X] : MetrizableSpace X <=> RegularSpace X /\ HasSigmaLocallyFiniteBase X := by sorry </pre>
4.4.8. The Bing Metrization Theorem. A topological space is metrizable if and only if it is regular and has a sigma-discrete base.	Topology	<pre> def IsDiscreteFamily {X : Type u} [TopologicalSpace X] {iota : Type v} (U : iota -> Set X) : Prop := forall x : X, exists V in nhds x, {i : iota (V inter U i).Nonempty}.Subsingleton def HasSigmaDiscreteBase (X : Type u) [TopologicalSpace X] : Prop := exists (iota : N -> Type v) (B : forall n, iota n -> Set X), IsTopologicalBasis {V : Set X exists n i, B n i = V} /\ forall n, IsDiscreteFamily (B n) theorem engelking_4_4_8_bing_metrization {X : Type u} [TopologicalSpace X] [T1Space X] : </pre>

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		<pre> MetrizableSpace X <=> RegularSpace X /\ HasSigmaDiscreteBase X := by sorry </pre>
5.1.9. Theorem. For every T1-space X the following conditions are equivalent: (i) the space X is paracompact; (ii) every open cover of the space X has a locally finite partition of unity subordinated to it; (iii) every open cover of X has a partition of unity subordinated to it.	Topology	<pre> def IsOpenCover {X : Type u} [TopologicalSpace X] {iota : Type v} (U : iota -> Set X) : Prop := (forall i, IsOpen (U i)) /\ union i, U i = univ theorem engelking_5_1_9_paracompact_partition_of_unity {X : Type u} [TopologicalSpace X] [T1Space X] : let locallyFinitePartition := forall (iota : Type v) (U : iota -> Set X), IsOpenCover U -> exists rho : PartitionOfUnity iota X, rho.IsSubordinate U let partition := forall (iota : Type v) (U : iota -> Set X), IsOpenCover U -> exists rho : iota -> C(X, R), 0 <= rho /\ (forall x, HasSum (fun i => rho i x) 1) /\ forall i, tsupport (rho i) subset U i List.TFAE [ParacompactSpace X, locallyFinitePartition, partition] := by sorry </pre>
5.1.38. The Tamano Theorem. For every Tychonoff space X the following conditions are equivalent: (i) the space X is paracompact; (ii) for every compactification cX of the space X the Cartesian product $X \times cX$ is normal; (iii) the Cartesian product $X \times \beta X$ is normal; (iv) there exists a compactification cX of X such that $X \times cX$ is normal.	Topology	<pre> def IsCompactification {X : Type u} {K : Type v} [TopologicalSpace X] [TopologicalSpace K] (e : X -> K) : Prop := IsEmbedding e /\ DenseRange e /\ IsCompact (univ : Set K) theorem engelking_5_1_38_tamano_theorem {X : Type u} [tX : TopologicalSpace X] [T35Space X] : let everyCompactification := forall (K : Type v) (tK : TopologicalSpace K) (e : X -> K), @IsCompactification X K tX tK e -> @NormalSpace (X x K) (tX.induced Prod.fst tK.induced Prod.snd) /\ T1Space (X x K) let someCompactification := exists (K : Type v) (tK : TopologicalSpace K) (e : X -> K), @IsCompactification X K tX tK e /\ @NormalSpace (X x K) (tX.induced Prod.fst tK.induced Prod.snd) /\ T1Space (X x K) List.TFAE [ParacompactSpace X, everyCompactification, NormalSpace (X x StoneCech X) /\ T1Space (X x StoneCech X), someCompactification] := by sorry </pre>
7.2.1. The Countable Sum Theorem. If a normal space X has a countable closed cover $\{F_j\}$ such that $\dim F_j \leq n$ for $j = 1, 2, \dots$, then $\dim X \leq n$.	Topology	<pre> def IsOpenCover {X : Type u} [TopologicalSpace X] {iota : Type v} (U : iota -> Set X) : Prop := (forall i, IsOpen (U i)) /\ union i, U i = univ def Refines {X : Type u} {iota : Type v} {kappa : Type w} (V : kappa -> Set X) (U : iota -> Set X) : Prop := forall j, exists i, V j subset U i def CoverOrderLE {X : Type u} {iota : Type v} (U : iota -> Set X) (n : N) : Prop := forall x : X, forall s : Finset iota, (forall i in s, x in U i) -> s.card <= n + 1 def CoveringDimensionLE {X : Type u} [TopologicalSpace X] (n : N) : Prop := forall (m : N) (U : Fin m -> Set X), IsOpenCover U -> exists (k : N) (V : Fin k -> Set X), IsOpenCover V /\ Refines V U /\ CoverOrderLE V n theorem engelking_7_2_1_countable_sum_theorem {X : Type u} [TopologicalSpace X] [NormalSpace X] {n : N} (hcover : exists F : N -> Set X, (forall j, IsClosed (F j)) /\ union j, F j = univ /\ forall j, CoveringDimensionLE (F j) n) : CoveringDimensionLE X n := by sorry </pre>
8.4.13. The Smirnov Theorem. Assigning to every compactification cX of a Tychonoff space X the proximity	Topology	<pre> def IsCompactification {X : Type u} {K : Type v} [TopologicalSpace X] [TopologicalSpace K] (e : X -> K) : Prop := </pre>

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delta(c) establishes a one-to-one correspondence between compactifications of X and proximities on X .

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IsEmbedding e /\ DenseRange e /\ IsCompact (univ : Set K)

def EquivalentCompactifications {X : Type u} {K L : Type v}
[TopologicalSpace X]
[TopologicalSpace K] [TopologicalSpace L] (e : X -> K) (f : X -> L) :
  Prop :=
exists h : K equiv L, h comp e = f

structure Proximity (X : Type u) where
  close : Set X -> Set X -> Prop
  empty_left : forall A, not close empty A
  intersects : forall A B, (A inter B).Nonempty -> close A B
  symmetric : forall A B, close A B <=> close B A
  union_left : forall A B C, close (A union B) C <=> close A C /\ close B C
  strong : forall A B, not close A B -> exists E : Set X, not close A E /\
not close E^c B

def IsAssignedProximity {X : Type u} {K : Type v} [TopologicalSpace X]
[TopologicalSpace K] (e : X -> K) (p : Proximity X) : Prop :=
forall A B : Set X,
  p.close A B <=> (closure (e '' A) inter closure (e '' B)).Nonempty

theorem engelking_8_4_13_smirnov_proximity_compactification
{X : Type u} [TopologicalSpace X] [T3Space X] :
(forall (K : Type v) (tK : TopologicalSpace K) (e : X -> K),
  @IsCompactification X K _ tK e ->
    exists p : Proximity X, @IsAssignedProximity X K _ tK e p) /\
(forall p : Proximity X, exists (K : Type v) (_ : TopologicalSpace K)
(e : X -> K),
  IsCompactification e /\ IsAssignedProximity e p) /\
forall (K L : Type v) (_ : TopologicalSpace K) (_ : TopologicalSpace L)
(e : X -> K) (f : X -> L) (p : Proximity X),
  IsCompactification e -> IsCompactification f ->
    IsAssignedProximity e p -> IsAssignedProximity f p ->
      EquivalentCompactifications e f := by
sorry

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